



A systematic literature review on risk perception of Artificial Narrow Intelligence

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ABSTRACT

The rise of artificial intelligence (AI) has captured global attention recently, prompting the need to understand how associated risks are perceived. This paper attempts to fill this gap by providing a comprehensive overview of the topic, focusing especially on the most problematic area of Artificial Narrow Intelligence. Utilising a systematic literature review, we examined 64 studies focusing on both statistical and qualitative aspects of AI risk perceptions. This research shows that current publications focus on Asia and North America, with the number of publications increasing significantly over the last three years. Research focuses primarily on three domains: Health, consumer behaviour and finance. This study has identified key factors that influence AI risk perceptions, including familiarity, trust and privacy, while also recognising confounding variables such as gender and political orientation. Still, one crucial shortcoming in existing literature emerges: While numerous studies examine how AI risk perceptions are conceived, there is a lack of systematic research on the formation of these perceptions.

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1. Introduction

The discourse on artificial intelligence (AI) is often accompanied by debates involving risk, where subjective fears typically interact with considerations about empirical evidence. A notable example is the public concern over AI in healthcare, where statistically AI systems may outperform human diagnostics in accuracy. (Bologheanu et al. 2023; Longoni, Bonezzi, and Morewedge 2020) This focus on potential hazards, highlights a well-known gap between statistical probability of adverse events and public perceptions of risk, generally characterised by beliefs rather than technical limitations. This literature review critically examines this dichotomy. The review uses an analytical frame that draws on the seminal work of Slovic (2016), Freudenburg (1988), and Schwing and Albers (2013) to conceptualise the divergence between ‘real’ and ‘perceived’ risk.

Risk perception research, particularly in the field of AI, is crucial for the development of effective risk communication strategies that enable informed decision-making.

The review answers the following questions: How has risk perception in the field of artificial intelligence evolved in the academic literature since 2000, and what are the prevailing themes and conclusions? The temporal limitation is set in recognition of the substantial advancements in AI capabilities and the potential shift in AI risk perceptions over time. In addition, it seems advisable to clearly delineate the concept of 'AI' for this study. In essence it describes an intelligent machine, mimicking human intelligence (McCarthy 2007).

In this endeavour, this analysis aims to thoroughly examine key areas, identify gaps in literature and detect trends in research, with a focus on the overarching concept of AI risk perception, focusing especially on the more problematic area of AI systems that are expected to have the most significant influence on the public's risk judgement. This broader perspective is essential for understanding the diverse and often conflicting views on AI. Although this approach may yield fewer results compared to specific application areas (e.g. self-driving cars), it remains crucial for a holistic understanding of AI risk perception.

2. Methodology

This paper serves two purposes: to provide a general overview of the current literature in the field of risk perception and AI, and to identify existing gaps that may be filled by future research.

2.1. Definitions

In this context, the term 'risk perception' requires definition, as it is a multi-layered concept that extends beyond communicating mere numbers and probabilities. The underlying assumption is that 'risk' and 'risk perception' are distinct aspects of reality (Rosa 1998; Aven and Renn 2009). Although there are contradictory opinions on the nature of risk – particularly with respect to cultural relativism (Douglas and Wildavsky 1983) – the more positivistic assumption would be that 'risk' represents an objective state of the world that exists independently of human perception and associated subjective assessments of the severity of hazards and their likelihood of occurrence, with 'risk perception' accounting for those by including, for example, fears, anxieties and reactions that may be based on other factors than objective risk. (Shrader-Frechette 1991; Rosa 1998; Jasanoff 1999; Pidgeon, Kasperson, and Slovic 2003; Aven and Renn 2009; Rosa 2010; Slovic 2016) Here, lay people in particular rely on their intuition when making risk assessments (Slovic 1987). While earlier authors assumed that the current risk level reflected the optimum of societal risk/benefit trade-offs, studies on risk perception not only showed that there are differences in lay people's and experts' risk perception, but also that lay people often were in disagreement of the current level of risk exposure; particularly for non-voluntary activities and technologies. (Fischhoff et al. 1978; Slovic 1987) For this paper it was decided to define the term in accordance with Rohrmann and Renn (2000). It thus '[...] refers to people's [subjective] judgement and evaluations of hazards they [...] are or might be exposed to' in contrast to 'results from quantitative risk assessment [...] or "predicted risk"' (Rohrmann and Renn 2000, 15).

The current advances in machine learning and neural networks require a distinct definition of AI. The initial challenge is to find a general and permanently valid definition of AI, as it presents a moving target that is constantly evolving and changing. Whereas some 25 years ago IBM's deep blue chess computer was considered the benchmark of AI, in the mid-2020s a standard smartphone could surpass its computing power. (Goodman and Keene 1997; Nilsson 2009; Russel and Norvig 2010; Puiu 2023) Known as AI effect, the definition of what is considered 'true intelligence' constantly changes as it is redefined upon successful completion of a previously defined task (Minsky 1985; Russel and Norvig 2010). Similarly difficult is the definition of the term 'intelligence' itself. A simple example illustrates this: it would hardly occur to anyone to claim that an elephant

is not intelligent just because it cannot play chess. (Brooks 1990; Nilsson 2009) Summarising various definitions of AI, two fundamental aspects emerge: the ability of machines to imitate human thinking and their rational behaviour (Russel and Norvig 2010; Haugeland 1985; Bellman 1978; Kurzweil 1990; Rich and Knight 1991; Charniak and McDermott 1985; Winston 1992; Poole, Mackworth, and Goebel 1998; Nilsson 1998). Lastly, the search for a standardised definition of AI is complicated by the multidisciplinary nature of AI, which spans from philosophy to computer science while touching on mathematics, economics, neuroscience, psychology, linguistics and others; all with their respective perspectives and definitional approaches. (Russel and Norvig 2010)

For these reasons, a broader definition has been chosen for this paper, which adopts the following segmentation: Artificial Narrow Intelligence (ANI) or Weak AI, Artificial General Intelligence (AGI) or Strong AI, and Artificial Super Intelligence (ASI). Current AI systems, which are assumed to have the most significant influence on the public's risk judgement, fall under the category of ANI. (Wilkinson 2001; Rizzi 2015) Despite recent technological advancements, marked by technologies such as OpenAI's ChatGPT and Alphabet's Deepmind, these systems do not yet qualify as AGI or ASI; they remain domain-specific and training-dependent. (IBM Cloud Education 2021; Naudé 2021; Greene 2022a; Greene 2022c; Greene 2022b; Kuusi and Heinonen 2022; Reed et al. 2022; Shah 2022) Although it is recognised that the boundaries between reality and science fiction can often blur in public perception, this study focuses on ANI and thus presumes that people's perception of risk is predominantly shaped by the current technological reality as opposed to a speculative vision of the future.

2.2. Scoping

The scoping study methodology was chosen as a guiding framework. It aims to capture the underlying key concepts and the main sources by reviewing a significant number of studies (Arksey and O'Malley 2005). The methodology combines the advantages of a systematic literature review, while allowing for different types of studies. Hence, it guarantees for a documented, evidence-based, reproducible, comprehensive, unbiased, and transparent approach of exploring the existing source situation (Cook, Mulrow, and Haynes 1997; Tranfield, Denyer, and Smart 2003; Arksey and O'Malley 2005; Weed 2005). The framework developed by Arksey and O'Malley (2005) is used. Besides selecting the research question, it consists of the following four steps.

2.2.1. Step 1 – systematic search query

A predefined search query with specific inclusion and exclusion criteria was performed in the following scientific databases: Clarivate, Web of Science, EBSCO Databases,¹ SCOPUS Elsevier. The queried articles had to fulfill the following criteria:

The articles had to...

- Be published between 2000 and 2024; if a time-bound search was possible. The search query in all databases was executed on February 24th, 2023.
- Be written in English.
- Be published in peer-reviewed journals or books (e. g. excluding conference papers).
- Match the following Boolean query for either subject terms, title, abstract, keywords, or keywords plus: (('risk perception' OR 'perceived risk') AND ('artificial intelligence' OR 'AI' OR 'Artificial Narrow Intelligence' OR 'ANI')).
- Exclude alternative AI abbreviations.²

2.2.2. Step 2 – initial result selection

Due to varying search engine designs, not all criteria could always be met. In such instances, the search was optimised for the most criteria while ensuring meaningful results. This meant

reducing search terms, not excluding certain terms, or having limited date restrictions. The exact query for each database is transparently provided in the attached dataset.

The list of articles underwent qualitative screening to eliminate obvious mismatches. Overall SCOPUS yielded the most extensive results. Articles were preliminarily assessed based on their abstracts, rating their topical fit on a five-point Likert-scale from 'None' to 'Very High'. Those rated 'Medium' or higher underwent further evaluation.

In principle, the preference was to err on the side of inclusivity, favouring more findings rather than omitting results.

2.2.3. Step 3 – in-depth screening

Subsequently, a detailed review of the full articles was conducted. All retained articles were accessed in full, and a binary selection ('in' or 'out') was applied. Articles were screened using the following criteria:

AI risk perception is...

- a central research focus (despite the initial review of the abstracts, several articles needed to be dismissed).
- thoroughly examined through quantitative or qualitative analysis (setting aside more journalistic or opinion pieces).
- studied within a software application context. As a result, articles with a focus on robotics or humanoid robots were intentionally excluded. This is in fact a controversial directional decision, as, for example, voice assistance systems, which also often appear in combination with hardware (e.g. Amazon Alexa or Google Nest), were still included. The rationale lies in the assumption that robots may trigger a distortion of risk perception due to their human-like appearance. Previous research has confirmed that by showing that the degree of anthropomorphism supports the emergence of uncanniness (S. Kim and McGill 2011; S. Y. Kim, Schmitt, and Thalmann 2019; Aubel, Pikturniene, and Joye 2022). Although risk behaviour was not directly influenced by anthropomorphism, uncanniness affected all four dimensions of risk perception (Aubel, Pikturniene, and Joye 2022). Yet, similar proclivities were found in this study; for AI-based hardware applications risk perception moderated purchase intention, which was positively influenced by cuteness (Yu et al. 2022).

2.2.4. Step 4 – extension of results

To further improve results, ten key journals in the fields of risk research, information technology, and behavioural science were hand-searched, with preference given to journals at the intersections of these fields. Complementary criteria were the appearance of the journal among the previous search results, prominence in the field, and a high h-index. On this basis 8 articles could be included in the result list; the details are reported in Table 1.

In addition, websites of renowned universities and prominent authors of (behavioural) risk research⁴ were reviewed without adding further articles.

Subsequently, an automated snowball method was applied by using the search engine 'Litmaps', using the citations of the search results so far as seed. This added two articles. In total, this added eleven more articles to the original 53 from the systematic search, providing a total of 64 sources. Details are provided in Table 2.

3. Results

For clearer insights, articles were organised, and key aspects summarised. This included findings on AI risk perception and framework data such as study population characteristics, observation location, AI subject area, data collection method, and study base model.

Table 1. Considered journals for the extension of the result list.

Index	Journal Name	ISSN	h-index ³	Initial search matches	Manually added articles	Articles in final selection
1	Computers in Human Behavior (UK)	07475632	226	8	1	4
2	Artificial Intelligence (NL)	00043702	161	1	-/-	-/-
3	Risk Analysis (UK)	15396924, 02724332	146	7	1	1
4	International Journal of Human-Computer Interaction (US)	15327590, 10447318	83	4	2	3
5	Journal of Risk and Uncertainty (NL)	15730476, 08955646	79	-/-	-/-	-/-
6	The Journal of Applied Behavioral Science (US)	15536879, 00218863	74	-/-	-/-	-/-
7	Journal of Risk Research (UK)	14664461, 13669877	74	3	2	2
8	Science and Engineering Ethics (NL)	13533452, 14715546	66	-/-	-/-	-/-
9	Perspectives on Behavior Science (CH)	25208969, 25208977	40	-/-	-/-	-/-
10	AI and Society (UK)	09515666, 14355655	39	9	2	5

Table 2. Overview of remaining results for the individual steps.

Database	STEP 1 – Number of results from initial search query	STEP 2 – Remaining articles after second screening	STEP 3 – Remaining articles after third screening	STEP 4 – Final Selection
Clarivate/ Web of Science	131	75	44	-/-
EBSCO	75	25	18	-/-
SCOPUS/ Elsevier	383	95	34	-/-
Added by the Author	-/-	-/-	-/-	11
Dublets	-120	-54	-43	-/-
Dismissed	-/-	328	77	-/-
Sum (excl. dublets)	469	141	53	64

The bold formatting highlights the result rows of the tables.

3.1. General overview and statistical insights

The final selection consists of 64 articles. Based on the original classification method for topical fit, 18 articles remained with a ‘very high’ score value, whilst 31 received a ‘high’ and 14 a ‘medium’ rating. It can thus be concluded that the original evaluation was comparatively error-prone, and the more detailed review was justified.

About one-third of the sample originates from seven scientific journals, with the publication ‘AI and Society’ as the most relevant single source with five articles. Based on the selection of journals with more than one relevant article this is also the journal with the lowest h-index and impact score (cf. Table 3). Examining the articles’ publication dates reveals a concentration between 2020 and 2023, with eight preceding this period, the earliest from 2013. This recency is both expected, given the topic’s heightened attention, and striking, considering AI classification discussions commenced as early as the late 1950s (McCarthy 2007).

The citation frequency reveals that Araujo et al. (2020) is the most cited article with 213 citations, twice as many as the next highest, Trivedi (2019).⁷ Given the recent spotlight on the topic, with most papers published in the last three years, it is unsurprising that over 60% of articles have under 10 citations.

Most study populations originate from the USA (14 articles) and China (12 articles), with notable contributions from Germany (four articles) and Japan (three). Continental analysis yields:

Table 3. Overview of the most relevant journals.

Index	Name of the Journal	<i>h</i> -index ⁵	Impact score ⁶	Number of articles included in review
1	AI and Society	33	2.87	5
2	Computers in Human Behavior	203	9.79	4
3	International Journal of Human-Computer Interaction	76	4.79	3
4	Frontiers in Psychology	133	3.88	2
4	International Journal of Bank Marketing	87	5.69	2
4	Journal of Medical Internet Research	158	7.52	2
4	Journal of Risk Research	65	5.97	2

The bold values indicate the lowest and highest value within the provided range.

Asia (22), North America (15), Europe (12), South America (one), and none from Oceania or Africa. Nine articles, analysing sources such as tweets, were broadly categorised as ‘worldwide’, suggesting an English bias yet lacking clear regional assignment. Five concept-driven articles remained unattributed. These patterns concur with the Global AI Index, emphasising US and Chinese dominance in the field of AI (Mousavizadeh, Mehta, and Darrah 2021).

Upon categorising the AI application fields from the articles, seven distinct areas emerged. ‘General’ AI research was the most prominent, encompassing 25 articles. ‘Shopping, Consumer, Household’ followed, accounting for nearly one-fifth of the content, primarily centred on chat and voice bots and virtual assistants – this focus was evident throughout the sample with 14 out of 64 articles. ‘Health’ covered close to 15%, emphasising Medical Imaging/Cancer Screening (three articles). ‘Banking/Finance’ represented 12.5%. ‘Cars and Transportation’ featured five articles, all on Autonomous Vehicles. Both ‘Education’ and ‘Tourism’ were addressed in two articles each. An overview is presented in Figure 1.

Additionally, papers were categorised by research method: qualitative (15%), mixed methods (5%), or quantitative (80%). Over 70% employed a survey study approach, exclusively quantitative, assessing variable relationships (Punch 2003). 15 studies referenced UTAUT⁸ or UTAUT^{2,9} frameworks explaining technology acceptance. Together with four studies on TAM¹⁰ they form

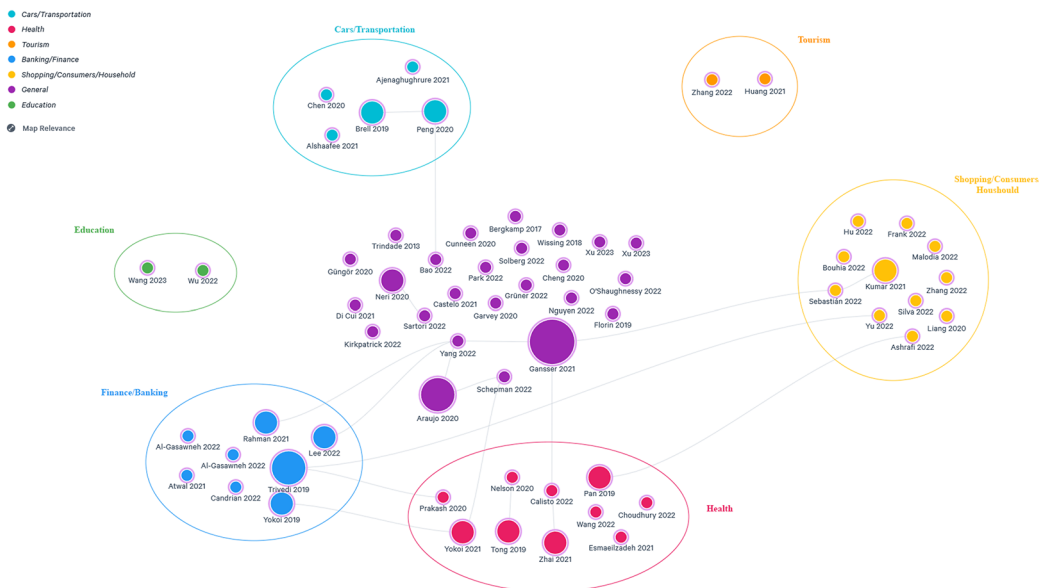


Figure 1. Allocation of articles to the fields of application and display of the cross-references between the papers. The size of the circles is representative for the relevance of the articles. The graphic was created with Litmaps (licensee thereof is the author).

the predominant theory models of the sample. In all but two instances, the models were modified, primarily extended, to better suit the papers' contexts.

For the articles that classified as quantitative research, study populations ranged from 25 to 3,682,015 subjects, with a median of 390. The 3.5 million figure pertains to a sentiment analysis of tweets about AI, exploring the role of experts in shaping AI perception (Neri and Cozman 2020). The largest human-centric study had 5,391 participants, examining the influence of AI narratives on perceptions (Sartori and Bocca 2022).

3.2. Findings from the analysis regarding risk perception

The following overview should be interpreted with due caution. While it provides valuable insights for future research, its generalisability may be limited. Tables 4–6 provide a brief overview of the factors that were found to influence AI risk perception and those that may promote intention to use AI. A distinction was drawn between factors that mitigate AI risk perception and factors that reinforce it. Limitations on the validity of certain factors are indicated where appropriate.

4. Discussion

4.1. Controversial influencing factors

First, the controversial influencing factors from Table 5 will be observed.

Gender – For nearly 30 years, studies examine gender differences in risk perception across various contexts and risks. (Flynn, Slovic, and Mertz 1994; Greenberg and Schneider 1995; Gustafson 1998; Schubert et al. 1999; Hitchcock 2001; Harris and Jenkins 2006; Olofsson and Rashid 2011; Brown, Largey, and McMullan 2021; Rodriguez-Besteiro et al. 2021) Quantitative evidence suggests females may exhibit higher risk sensitivity, yet this effect is inconsistent and context-dependent (Schubert et al. 1999; Hitchcock 2001). The examined studies have examined a gender effect across quite different use cases (cf. Table 5), complicating clear attribution of observed effects to AI usage or broader risk assessment patterns. Hence, further in-depth, contextual analysis seems essential before affirming gender's influence on AI risk perception. Nonetheless, it is advised that future studies consider gender as a distinct variable.

Political and Religious Orientation – An initial perspective might suggest that conservative worldviews tend to emphasise the status quo and are therefore less receptive to new technologies. (Jost et al. 2003) This assumption is echoed in two of the reviewed studies. (Castelo and Ward 2021; Park et al. 2022) Yet, U.S. AI legislative initiatives emerged during a conservative administration. (Hutson 2019; Metz 2019; *Exec. Order No. 13859 - Maintaining American Leadership in Artificial Intelligence* 2019; National Artificial Intelligence Initiative Act of 2020 2021) The study of Bao et al. among 2,700 U.S. adults also found no statistically significant differences in risk benefit perceptions regarding AI, based on political ideology or religious (2022). Similarly, Gruener confirmed the marginal relevance of political orientation for AI within a German context (2022). Disparate findings, combined with the often-contradictory nature of policy debates on AI, prompt an in-depth examination of how opinions on AI are formed. Other influential factors, such as trust in big corporations or governments, raise the question of potential multicollinearity in the understanding of AI perceptions. (Tong and Sopory 2019; O'Shaughnessy et al. 2022; Park et al. 2022; Schepman and Rodway 2022) Although this review cannot confirm such relationships, it does point towards compelling lines for future research.

Age – A similar chain of reasoning may be applied to the finding that negative attitudes towards AI increase with age, which is the reason why this factor was included in Table 5. This classification seems appropriate in light of the fact that existing literature is not clear regarding the link between risk perception and age. Whilst some authors posit older adults

Table 4. Overview of influencing factors for risk perception regarding artificial intelligence.

Influencing Factors on AI Risk Perception (largely consistent results)			
Influencing Factor	Positive Influence, (i. e. lowers AI risk perception)	Negative Influence, (i. e. increases AI risk perception)	Application Area(s) for which factor appeared
Familiarity and Personal Innovativeness	• Prior exposure to AI positively correlates with enhanced acceptance and favourable perception.(Brell, Philipsen, and Ziefle 2019; Tong and Sopory 2019; Vimalkumar et al. 2021; Park et al. 2022; Rohden and Zeferino 2022; Sartori and Bocca 2022; Schepman and Rodway 2022; Ashrafi and Easmin 2023) • In intelligent information systems: positive technology perception positively influences intention to adopt. (Park et al. 2022) • In AI medical imaging: moderate subjective and objective knowledge correlates with reduced negative attitudes and increased usage intention. (Wang et al. 2022)	• In health: prior experience and knowledge inversely affect risk perception for both clinicians and non-clinicians. (Pan et al. 2019; Wang et al. 2022)	• General • Health • Shopping/Consumers/ Household • Cars/Transportation
Trust	• Increased experience with driving technologies enhanced benefit perception but not barrier perception. (Brell, Philipsen, and Ziefle 2019) • Trust in AI mitigates perceived risk. (Ajenaghughure, Sousa, and Lamas 2021; Choudhury 2022; Rohden and Zeferino 2022; Silva et al. 2022) • Trust in government positively correlates with technology acceptance. (Park et al. 2022)	• In health: AI is less trusted than human doctors, even with superior performance, especially among chronic patients. (Esmailzadeh, Mirzaei, and Dharanikota 2021; Yokoi et al. 2021) • High corporate distrust negatively influences attitudes towards AI. (Schepman and Rodway 2022) • Conservatism decreases trust in AI. (Castelo and Ward 2021) • Psychopathy and Machiavellianism correlate with limited risk perception. (Wissing and Reinhard 2018)	• General • Banking/Finance • Cars/Transportation • Health • Shopping/Consumers/ Household • General
Personality	• Introverted traits and heightened general correlate with more positive attitudes towards AI. (Schepman and Rodway 2022)	• In skin cancer screening: reduced positive sentiment towards AI correlate with heightened AI risk perceptions and reduced AI benefit evaluations. Uncertain risk messages intensified risk perceptions and benefit evaluation, compared to low-risk messages.(Tong and Sopory 2019)	• Health • Shopping/Consumers/ Household
Affect	• -/-	• Negative valence (i. e. emotional value or positive/negative charge) increases perceived risk, but positive valence did not reduce it. (Frank, Chrysochou, and Mitkidis 2022)	• Shopping/Consumers/ Household
Power	• In voice shopping: Perceived power over AI assistants mitigates users' risk perceptions. It was shown that power experience could attenuate users' risk perceptions. (Hu, Lu, and Wang 2022)	• -/-	• Shopping/Consumers/ Household
Communication and Influencer Endorsement	• Word of mouth content may mitigate consumer risk perception. (Zhang, Sun, et al. 2022) • In Banking/Finance: influencer endorsement moderates risk perception and increases usage intention. (Al-Gasawneh, Al-Hawamleh, et al. 2022; Al-Gasawneh, Al-Okdeh, et al. 2022)	• -/-	• Banking/Finance • Shopping/Consumers/ Household
Transparency	• Transparency in AI decisions-making reduces risk perception. (Esmailzadeh, Mirzaei, and Dharanikota 2021; Candrian and Scherer 2022)	• -/-	• Health • Banking/Finance
Privacy	• In e-commerce: heightened security perception for private data decreases risk perception for financial AI-services. (Al-Gasawneh, Al-Hawamleh, et al. 2022)	• -/-	• Banking/Finance

Table 5. Overview of controversial influencing factors for risk perception regarding artificial intelligence.

Potentially Influencing Factors on AI Risk Perception (partly contradictory results)			
Influencing Factor	Positive Influence, (i. e. lowers AI risk perception)	Negative Influence, (i. e. increases AI risk perception)	Application Area(s) for which factor appeared
Gender RESULTS INCONSISTENT	• -/- Gender did not moderate risk perception. (Araujo et al. 2020; Bouhia et al. 2022; Zhang, Zhu, et al. 2022)	• Female gender is associated with more negative attitudes towards AI. (Chen et al. 2020; Cheng and Jiang 2020; Castelo and Ward 2021; Sartori and Bocca 2022; Schepman and Rodway 2022) • For chatbots: men perceive higher risks than women. (Bouhia et al. 2022)	• General • Shopping/Consumers/ Household • Cars/Transportation
Political and Religious Orientation RESULTS INCONSISTENT	• -/- Differences in political ideology or religion were not found to be statistically relevant. (Bao et al. 2022; Gruener 2022)	• Conservatism and aversion of liberal corporations enhance risk perception (Castelo and Ward 2021; Park et al. 2022) • Religiosity heightens risk perception (Park et al. 2022)	• General
Age	• -/-	• Greater age correlates with more negative attitudes towards AI. (Araujo et al. 2020; Chen et al. 2020; Cheng and Jiang 2020; Castelo and Ward 2021; Sartori and Bocca 2022; Schepman and Rodway 2022)	• General • Cars/Transportation

may perceive less personal risk due to socioemotional selectivity theory, others argue for an age-associated decline in risk tolerance. (Hariharan, Chapman, and Domian 2000; Löckenhoff and Carstensen 2007) With regard to technology, it has been found that the reduction of psychological barriers may foster acceptance of older individuals. (Adams, Stubbs, and Woods 2005) The findings of this paper in relation to AI seem to support that (see ‘Familiarity and Personal Innovativeness’ in Table 4).

4.2. Delimitation of risk (perception) and trust

Thus far the predominant focus centred on risk perception. This raises the question of whether the analysis might benefit from including ‘trust’ as part of the search query, given its prominent appearance as one of the central influencing factors in Table 4.

A word cloud analysis¹² of the examined articles further highlights the prominence of ‘risk’ and ‘trust’ (Figure 2).

First, the intricate interrelationship between risk perception and trust resembles a complex dichotomy, with many studies indicating a significant bidirectional correlation, whereby their interdependence should not be understood as absolute, but rather as a mutual bidirectional relationship (Rousseau et al. 1998; Eiser, Miles, and Frewer 2002; Siegrist, Gutscher, and Earle 2005; Siegrist 2021).

This is affirmed by works such as Vimalkumar et al. (2021) and Park et al. (2022), which extended the UTAUT2 model for technology acceptance and established a strong correlation between risk perception and trust in the context of AI. Nevertheless, there is a firm consensus distinguishing risk perception and trust as separate constructs (Slovic 1987; Siegrist and Cvetkovich 2000; Sjöberg 2001; Eiser, Miles, and Frewer 2002). Examining the objective of trust

Use of Technology			
Influencing Factor	Positive Influence, (i. e. lowers likelihood of AI use)	Negative Influence, (i. e. increases likelihood of AI use)	Application Area(s) for which factor appeared
Risk Perception RESULTS INCONSISTENT	- Risk Perception does not influence technology use; users appear largely undeterred by perceived risk. (Trivedi 2019; Zhai et al. 2021) - The practical impact of negative influence is minimal. (Rahman et al. 2021)	In Mobility, Household, and Banking (excl. Health): risk perception negatively impacts behavioural intent. (Atwal and Bryson 2021; Gansser and Reich 2021; Rahman et al. 2021)	- General - Health - Banking/Finance - Cars/Transportation
Privacy Risk RESULTS INCONSISTENT	In virtual assistants: perceived privacy risks do not influence user intent to use. (García de Blanes Sebastián, Sarmiento Guede, and Antonovica 2022)	In chatbots: Privacy risks negatively impact continued use and customer satisfaction. (Cheng and Jiang 2020; Zhang, Zhu, et al. 2022; Ashrafi and Easmin 2023)	- General - Shopping/Consumers/ - Household - Tourism
Usability	Facilitating conditions positively influence technology acceptance. (Park et al. 2022)	Negative attitudes intensify with perceived usability challenges. (Park et al. 2022)	General
Time risk ¹¹	-/-	In chatbots for tourism: Time risk negatively influenced customers' intent to use. (Zhang, Zhu, et al. 2022)	Tourism
Chronic Disease	-/-	Patients with chronic disease show lower propensity to rely on AI (even under physician supervision). (Esmaeilzadeh, Mirzaei, and Dharanikota 2021)	Health
Value similarity RESULTS INCONSISTENT	Shared investment strategy had marginal significance on AI adoption. (Yokoi and Nakayachi 2019)	-/-	Banking/Finance
Value similarity does not bolster trust towards AI. (Yokoi and Nakayachi 2019)			



Figure 2. Word cloud from the text of the analysed articles.

and risk perception regarding human behaviour and decision making, it appears they touch on different strands:

- Risk Perception encompasses intuitive evaluations of potential hazards influenced by a multitude of factors such as personal experience, knowledge, emotions, cultural norms, and also trust. (Slovic 1987; Siegrist, Gutscher, and Earle 2005) For instance, while one individual might deem skydiving as risky due to injury potential, another might see it as thrilling and secure.
- Trust, a multifaceted construct, may vary in its dimensions and measurements across different studies (Siegrist, Gutscher, and Earle 2005). It serves as a mechanism to navigate complex environments and as societal lubricant, relying on reciprocal dependability for collective success (Siegrist 2021). Evolving over time based on past experiences, trust undergoes three phases: building, stability, and dissolution. (Rousseau et al. 1998)

Though intertwined, risk perception and trust remain distinct constructs influencing different types of decisions. Despite trusting expertise or skill in perceived risky activities like skydiving, one might still recognise inherent risks. Conversely, trust does not negate perceived risks in certain actions or decisions. The precise demarcation of these subjects justifies the focused search queries, acknowledging that not all authors differentiate them distinctly—a trend observed in some articles within this review.

4.3. Suspicion of AI-stigmatization

Although artificial intelligence has been gradually introduced over the years, many people have had little exposure to it, which could change given the introduction of ChatGPT and other AI tools with much simpler user interfaces. Results from Table 4 indicate that, except in the health sector, prior AI experience correlates with a more positive risk perception. This contrasts with earlier literature on information technology innovations adoption, where the *novelty* aspect often skewed benefit perceptions that eclipsed the risks or led to higher concerns about established risks. (Wells et al. 2010; Etienne et al. 2018) Literature suggests those familiar with AI might transcend initial affective reactions, leading to more deliberate risk assessments. (Skagerlund et al. 2020) Given that negative valence can also strongly impact risk perception, the question arises whether AI faces stigmatisation, supported for example by the *Terminator Syndrome*¹³ or that risk perception of AI moderates positive influencing factors, such as cuteness. (Slovic 2016, 341–346; Frank, Chrysochou, and Mitkidis 2022; Yu et al. 2022) Advanced technologies often encounter stigmatisation, paralleled by public ambiguity. (Slovic 2016, 341–346) The mixed results of this review could support this hypothesis, were it not for the predominantly positive reporting on AI. (Garvey and Maskal 2020; Neri and Cozman 2020)

Hence, although there is a possibility for that assumption, only initial indicators support it and comprehensive investigation is necessary. The results from Table 4 further imply the need to differentiate findings based on application domains.

4.4. The role of the News Media and implications for public policy

News Media's coverage of AI is often perceived as negative, amplifying a generally negative public perception of AI (Shead 2018; Garvey and Maskal 2020). This literature review suggests that this position lacks generalisability. Three papers have examined the influence of the News Media in more detail. The results show that while discourse has become more risk-aware, it is incorrect that a uniformly negative narrative is presented. (Garvey and Maskal 2020; Cui and

Wu 2021; Cui and Wu 2021; Nguyen and Hekman 2022) Notably, a Chinese study found no significant correlation between media use and risk perception of AI (Cui and Wu 2021). Despite the caveat of China's media landscape, the study challenges the simplistic notion that negative news directly heightens risk perception among lay people. While AI coverage has grown, with a predominantly positive sentiment, the *Terminator Syndrome* persists. (Garvey and Maskal 2020; Neri and Cozman 2020) It can be described most simply with the imagery of the movie 'Terminator': artificial intelligence waging war against humanity and posing an existential threat to it. This prevents broader public discourse before it has even begun. (Garvey and Maskal 2020) This finding is all the more remarkable given that few experts and influencer endorsement seem to have notable influence on the formation of public opinion (Neri and Cozman 2020; Al-Gasawneh, Al-Hawamleh, et al. 2022; Al-Gasawneh, Al-Okdeh, et al. 2022; Zhang, Zhu, et al. 2022).

The challenge arises in broadening public discourse on AI without undue bias. This concern deepens considering Bao et al.'s findings, where the 'ambiguous' and 'indifferent' classes expressed low agreement that the public should have a say in scientific developments. Which is all the more remarkable when considering that the most vulnerable to being disadvantaged are among the largest proportions of these classes: racial minorities, women, and less educated people. (Eubanks 2018; Bao et al. 2022) In parallel, trust in government correlates positively with technology acceptance, while individuals who believe in conspiracy theories seem to be more susceptible to misinformation, which further complicates the design of adequate AI risk communication. (Gruener 2022; Park et al. 2022) Recognising this, constitutional democracies are confronted with a lack of proper instruments for societal risk management of AI. Applying Beck's risk society to AI, for example, would suggest to treating perceived risks as actual risks, which inevitably would lead to the over-regulation of non-existent or exaggerated phantom risks, which could jeopardise effective risk management. (Bergkamp 2017) The need for further research in this domain is evident. Simultaneously, it should be clear that while efforts are underway to incorporate risk perception into algorithms, deeper exploration of the mental models shaping risk perceptions of AI is vital. (Trindade and Antunes 2013)

4.5. General remarks on the results

To better interpret the results, it is useful to recall three aspects:

1. Most studies focused on distinct AI use cases (e.g. chatbots, autonomous vehicles) limiting generalisability. One should be aware of this constraint.
2. Cultural nuances significantly influence the results, as highlighted before. Variances in study populations necessitate caution when extrapolating results across linguistic and cultural boundaries (Slovic 2016; Spencer 2016).
3. Risk perception is deeply influenced by a study's population. Differences in risk perceptions, notably between experts and the public, have been well-documented (Slovic 1987; Slovic 2013; Slovic 2016; Johnson and Slovic 1996; Morgan et al. 2001; Savadori et al. 2004; Lundgren, McMakin, and Wiley 2018). Within this study's context, two papers particularly reflect this variance: O'Shaughnessy et al. (2022) investigates gaps in perception, while Neri and Cozman (2020) highlight experts' influential power.

Furthermore, many confuse risk with the intention to adopt certain technologies or engage in certain behaviour. While risk perception influences behaviour, it is not synonymous with outright rejection. (Slovic 1987; Slovic 2013; Slovic 2016; Rohrmann and Renn 2000) The data from Table 6 reflect great variability in risk perception according to the different application areas of AI, highlighting the need for concise demarcation in future research. Neither a broad perspective on AI's overarching risk perception nor a niche focus on specific technologies appears

very effective. While individuals might distinguish risk perceptions across sectors like banking and health, laypeople likely struggle to discern nuances in technological approaches and their inherent risks.

Although numerous studies mention the term risk perception, few defined it. This is reflected in similar methodologies and models for technology acceptance that were used to capture risk perception, predominantly relying on surveys and UTAUT2. While surveys are a standard tool for assessing risk perception, the widespread use of rather basic Likert scales directly polling risk perception stood out. Although this approach also yields insights, few captured the multidimensionality of AI risk perception through a more nuanced approach, as is often the case in other areas; e.g. smoking. (Kaufman et al. 2020) Hence, risk perception was often reduced to data protection concerns. It is therefore tempting to assume that the study of risk perception was in many cases a bandwagon effect, while the main focus was on other objects of study. Further support comes from the fact that it was often a minor research hypothesis. Only limited studies investigated the reasons for specific risk perceptions related to AI. It is essential to contextualise what may seem to be exhaustive results, without casting aspersions on the authors. Consequently, fewer findings emerge, in a topic area whose relevance is emphasised by almost all authors.

5. Conclusion

Overall, the results were much less conclusive than might have been expected given the number of studies. A deeper exploration into the determinants of human risk perception concerning artificial intelligence is merited. Delving into the underlying mental models could enrich future research, especially as no systematic studies on these models were identified in this analysis. Such studies could enhance the specificity of risk perception concerning AI, reducing its often-generalised nature. However, future research should limit itself to specific application areas of AI to ensure transferability of results.

6. Notes, limitations, and directions for future research

At the time of writing this manuscript, Chat-GPT had been introduced to the public. With the help of a language processing model, users can communicate with the integrated algorithm via chat and search for answers on various topics. This has significantly spiked interest in AI. While the trends identified in this review remain relevant, it is possible that risk perceptions have changed considering this development.

One anecdote deserves mention: For this paper using Chat-GPT was considered. The tool was employed to extract key information from texts through chat queries. However, the technology's limitations became apparent as it often produced erroneous results, despite its consistent response generation. Although parts of the answers were correct, identifying errors necessitated a detailed review of the original texts. This experience underscored the risks of blind trust in AI.

Notes

1. This included: Business Source Complete, CINAHL, EconLit, ERIC, GreenFILE, Library, information Science & Technology Abstracts, APA PsycArticles, APA PsycInfo, Regional Business News, SocINDEX, Psychology and Behavioral Science Collection, MEDLINE, APA PsycBooks, EBSCOhost (eBook Collection, Audiobook Collection, eBook Academic Collection, eBook Open Access Collection), OpenDissertations, Health and Psychsocial Instruments, MLA Directory of Periodicals, MLA International Bibliography.
2. Several trial runs of the search indicated 'avian influenza', 'anal intercourse', 'american indian' as the most common false positive findings.

3. The *h*-index indicates the number of journal articles (*h*) cited at least *h* times. As per SCImagojr.com, dated 05/11/2023 (with 2022 as base year).
4. In Detail: Publications of Gisela Böhm from Bergen University, Publications of Ragnar Löfstedt from King's College London, Centre for Research Ethics and Bioethics (CRB) at Uppsala University, Center for Interdisciplinary Risk and Innovation Studies (ZIRIUS) at University of Stuttgart and Harding Center for Risk Literacy at University of Potsdam, and Research Institute for Sustainability Helmholtz Centre Potsdam as well as Publications of Ortwin Renn (ibid).
5. Ibid.
6. The impact score (also Journal impact score) is a measure of the yearly average number of citations to recent articles published in that journal. It is based on Scopus data. The higher the IS of a journal, the better it is ranked.
7. As per Litmaps.com, dated 06/02/2023.
8. UTAUT: Unified Theory of Acceptance and Use of Technology (Venkatesh et al. 2003).
9. UTAUT2: Based on UTAUT, the refined model aims to emphasize the consumers' context of use (Venkatesh, Thong, and Xu 2012).
10. TAM: Technology Acceptance Model.
11. 'Time risk refers to the risk that a person will waste time and efforts when using a technology.' (Zhang, Zhu, et al. 2022).
12. The selected texts were processed using Python (v 3.10) with the 'PyPDF4' package. Words were cross-referenced with the NLTK English corpus (236,736 words as of May 3, 2023). Non-matching words and predefined stop words were excluded, with additional manual omissions (e.g., 'number', 'study', etc.). A 550-word cloud was crafted using the 'wordcloud' and 'matplotlib' packages.
13. Further elaborated in the following subchapter.

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